UPSC Prelims

Science & Tech

Class No. 5

Reserve class (biotechnology)

Recap

- Innate and Acquired Immunity
- B cells (humoral immunity) and T cells (cell mediated immunity)
- Active, Passive and Auto-Immunity
- Principle of Memory for Vaccination
- Anti- gen vs Patho – gen
- Active and Passive Vaccination

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AEFI : Adverse Effects Following Immunization

- There is no such thing as a "perfect" vaccine which protects everyone who receives it AND is entirely safe for everyone.
- Effective vaccines (i.e. vaccines inducing protective immunity) may produce some undesirable side effects which are mostly mild and clear up quickly.
- EFIs can be related to the vaccine itself (product or quality defect-related reactions), to the vaccination process (error or stress related reactions) or can occur independently from vaccination (coincidental) and are classified as:
- Vaccine product-related reaction
- Vaccine quality defect-related reaction
- Immunization error-related reaction
- Immunization anxiety-related reaction
- Coincidental event

Types of Vaccines

- Based on method of Administration
- Im
- Iv
- Sc
- Nasal
- Oral

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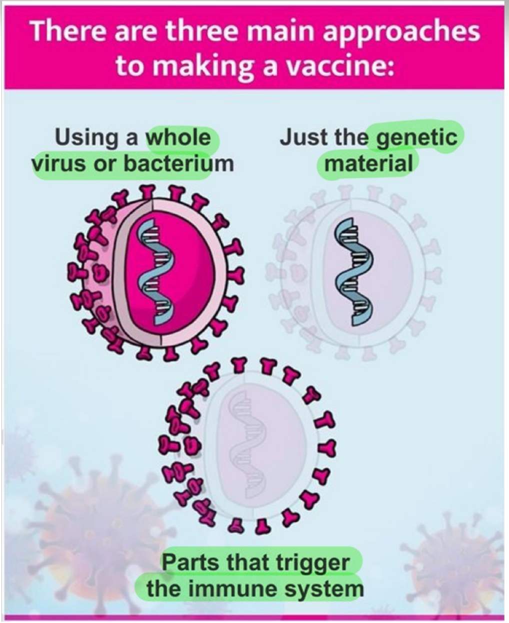
Types of vaccines : based on pathogen

- Live Attenuated
- Inactivated
- Subunit
- Toxoid
- Recombinant
- mRNA
- DNA



Inactivated Vaccine

- Inactivated vaccines use the killed version of the germ that causes a disease.
- Inactivated vaccines usually don't provide immunity (protection) that's as strong as live vaccines. So you may need several doses over time (booster shots) in order to get ongoing immunity against diseases.
- Inactivated vaccines are used to protect against:
- Hepatitis A
- Flu (shot only)
- Polio (shot only)
- Rabies

Live Attenuated Vaccine

- Live vaccines use a weakened (or attenuated) form of the germ that causes a disease. Because these vaccines are so **similar to the natural infection** that they help prevent, they create a strong and **long-lasting immune response**. Just 1 or 2 doses of most live vaccines can give you a lifetime of protection against a germ and the disease it causes.
- **Limitations** :
- Because they contain a small amount of the weakened live virus, they may cause disease in some people, especially those with a **compromised immune system**
- They need to be kept cool, so they don't **travel** well.
- Live vaccines are used to protect against:
- Measles, mumps, rubella (MMR combined vaccine)
- Rotavirus, Smallpox, Chickenpox, Yellow fever

Subunit Vaccine

- A subunit vaccine is a vaccine that contains purified parts of the pathogen that are antigenic, or necessary to elicit a protective immune response.
- A "subunit" vaccine **doesn't contain the whole pathogen**, unlike live attenuated or inactivated vaccine, but contains **only the antigenic parts** such as proteins, polysaccharides or peptides.
- Because the vaccine **doesn't contain "live" components** of the pathogen, there is no risk of introducing the disease, and is safer and more stable than vaccine containing whole pathogens. Other advantages include being well-established technology and being **suitable for immunocompromised individuals**
- Disadvantages include being relatively complex to manufacture compared to some vaccines, possibly requiring adjuvants and booster shots, and requiring time to examine which antigenic combinations may work best.

Recombinant Vector Vaccine

- Uses a safe virus or bacteria to deliver specific sub parts of a pathogen.
- Ex : Spike protein of coronavirus
- Immune response is generated without causing a disease
- Genetic instructions of making such a protein are inserted into a safe virus (through RDnaTech)
- Several different viruses have been used as vectors, including influenza, vesicular stomatitis virus (VSV), measles virus, and adenovirus, which causes the common cold. Adenovirus is one of the viral vectors used in some COVID-19 vaccines being studied in clinical trials.
- Viral Vector vaccines produce immunity from both B and T cells (how? Not important)

Conjugate vaccine

- A conjugate vaccine is a type of subunit vaccine which combines a weak antigen with a strong antigen as a carrier so that the immune system has a stronger response to the weak antigen.
- Most vaccines contain a single antigen that the body will recognize. However, the antigen of some pathogens does not elicit a strong response from the immune system, so a vaccination against this weak antigen would not protect the person later in life.
- In this case, a conjugate vaccine is used in order to invoke an immune system response against the weak antigen. In a conjugate vaccine, the weak antigen is covalently attached to a strong antigen, thereby eliciting a stronger immunological response to the weak antigen.
- Most commonly, the weak antigen is a polysaccharide that is attached to strong protein antigen. However, peptide/protein and protein/protein conjugates have also been developed.

Polysaccharide Vaccine

- Polysaccharides : poly carbohydrates
- Vaccines that consist of Poly carbohydrate molecule that generates immune response
- Generally vaccines are made of protein, that's why polysaccharide vaccines are special.

Nucleic Acid Vaccines

- Using genetic material from a pathogen to stimulate an immune response.
- Could be Using DNA (dna vaccine) or RNA (rna vaccine).

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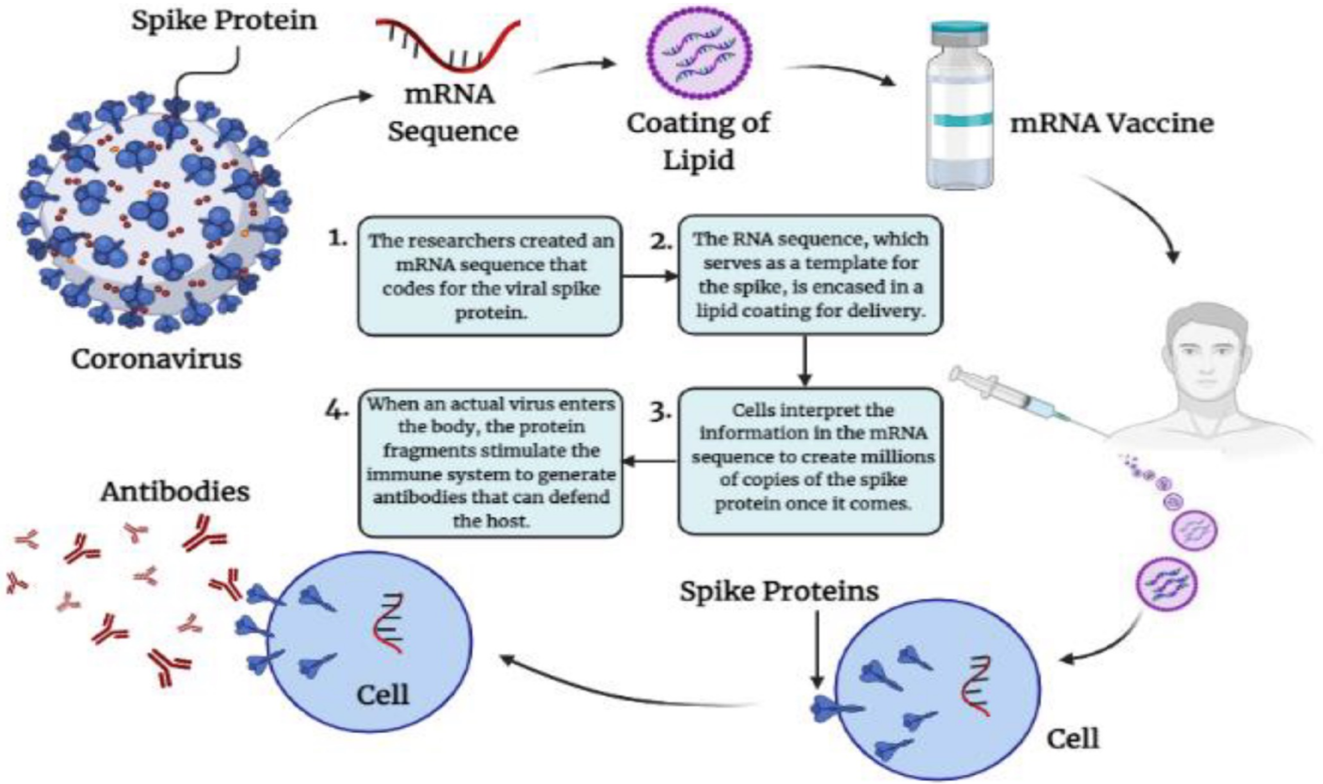
mRNA vaccine

- What is mRNA?
- What is the vaccine principle?
- Now you try and tell me what a mRNA vaccine will do.

mRNA Vaccines : type of nucleic acid vaccine

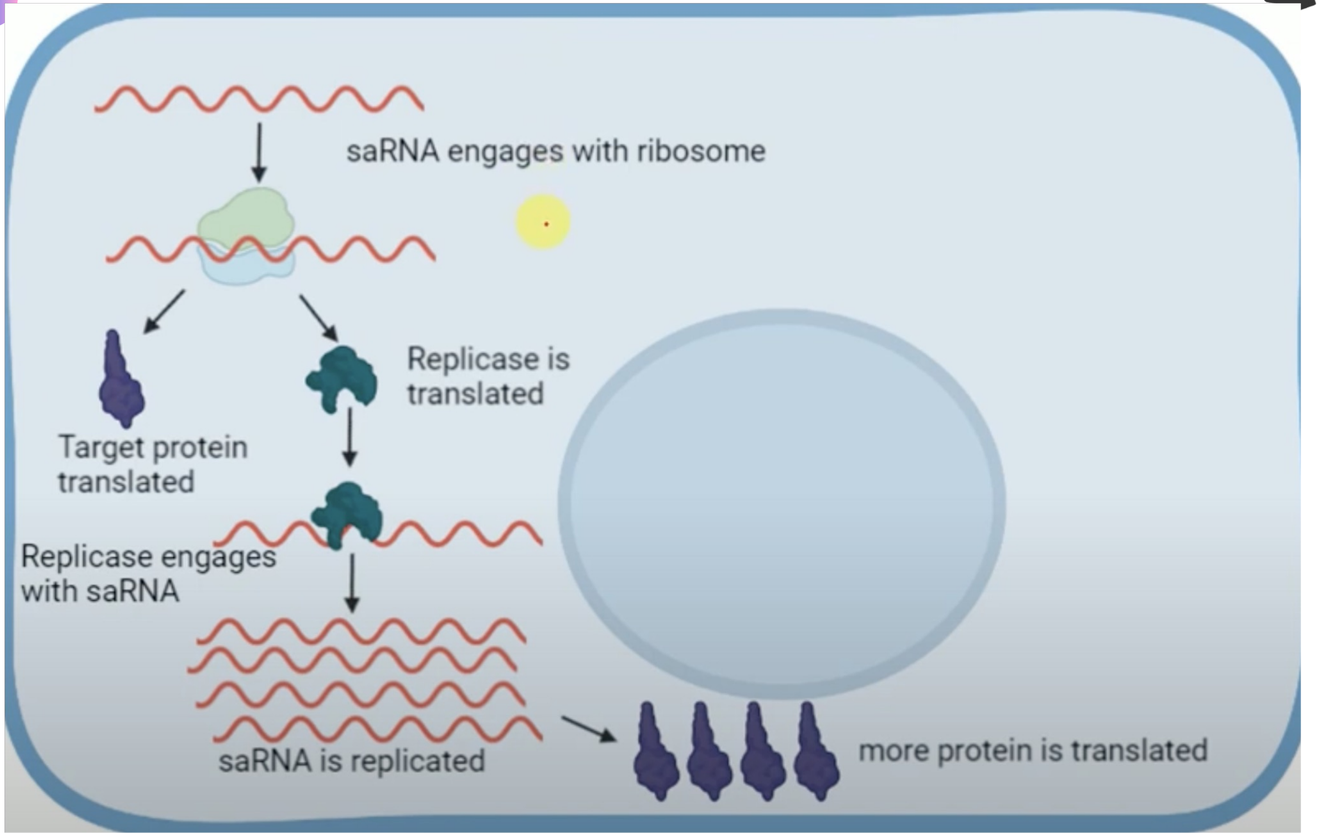
- An mRNA vaccine is a type of vaccine that uses a copy of a molecule called messenger RNA (mRNA) to produce an immune response. The vaccine delivers molecules of antigen-encoding mRNA into immune cells, which use the designed mRNA as a blueprint to build foreign protein that would normally be produced by a pathogen (such as a virus) or by a cancer cell. These protein molecules stimulate an adaptive immune response that teaches the body to identify and destroy the corresponding pathogen or cancer cells.


Spike Protein
mRNA
Sequence
Coating of
Lipid
mRNA Vaccine
Coronavirus
1.
The researchers created an mRNA sequence that codes for the viral spike protein.
2.
The RNA sequence, which serves as a template for the spike, is encased in a lipid coating for delivery.
3.
Cells interpret the information in the mRNA sequence to create millions of copies of the spike protein once it comes.
4.
When an actual virus enters the body, the protein fragments stimulate the immune system to generate antibodies that can defend the host.
Antibodies
Cell
Spike Proteins
Cell

New Type : Self Amplifying mRNA vaccine

- A Self-amplifying messenger RNA (mRNA) vaccine, ARCT-154, developed by Arcturus Therapeutics Holdings (US), showed promising results against Covid- 19 .
- Conventional mRNA : codes for spike protein of Covid-19 virus.
- Self amplifying mRNA vaccine codes for 4 additional proteins that enable amplification of the mRNA strand.
- Stronger Immunity, Storage is easier, minimises dose of mRNA, cost reduced.

Self Amplifying mRNA

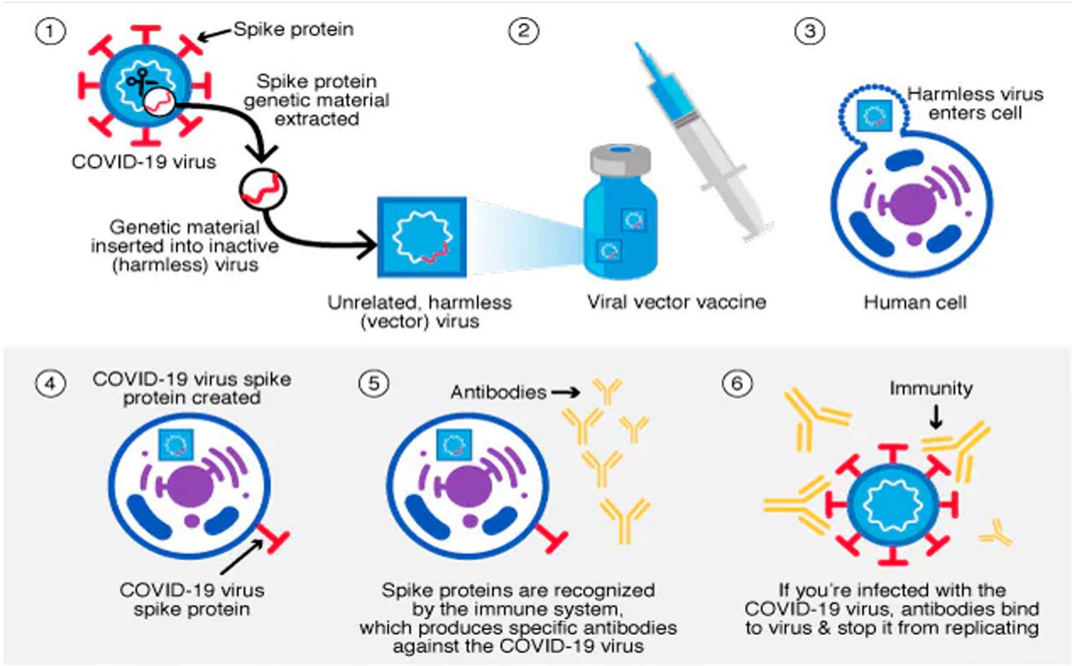

saRNA engages with ribosome
Replicase is translated
Target protein translated
Replicase engages with saRNA
saRNA is replicated
more protein is translated

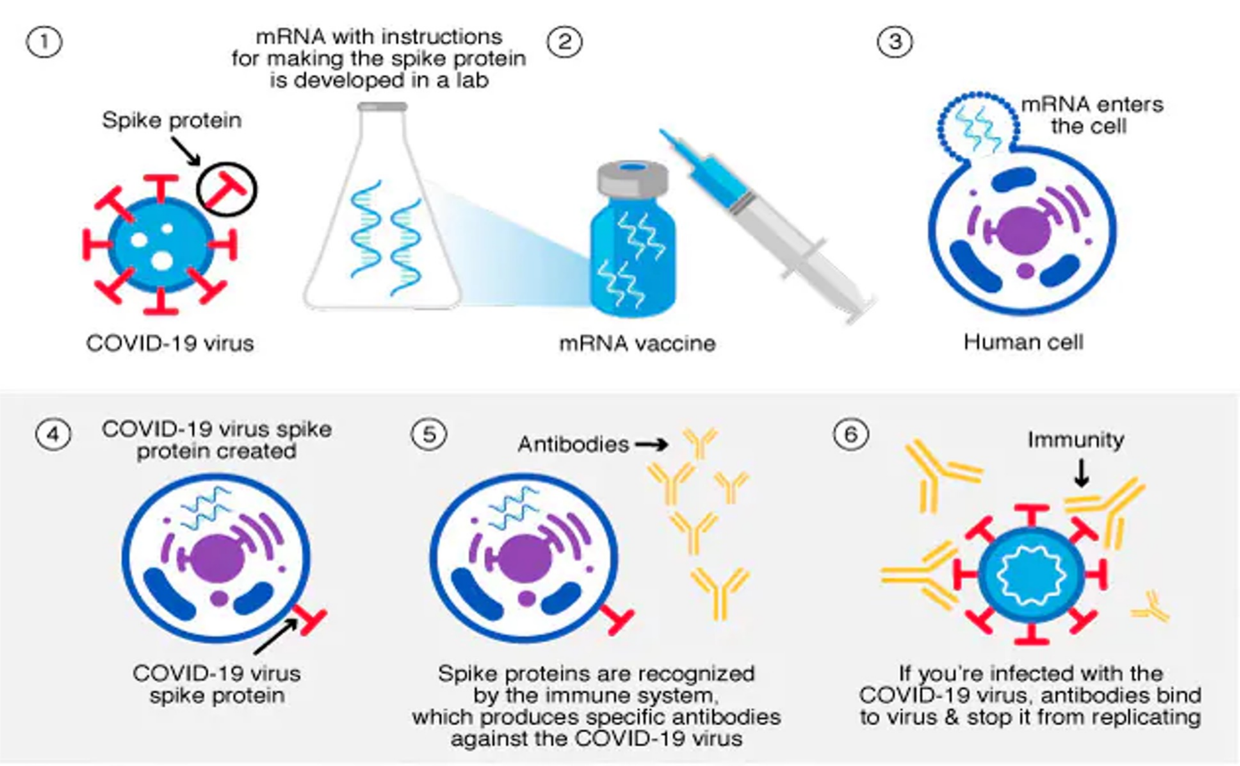
DNA Vaccines

- Piece of DNA encoding the antigen is inserted into the bacterial plasmid.
- DNA plasmids are inserted intra muscularly and driven into cells with help of electroporation.
- DNA based vaccines are more stable : don’t require ultra cold storage so they are more cost effective.
- Video : How to DNA vaccines work (Elsevier)

Covid 19 Vaccines

- mRNA
- Viral Vector
- Subunit
- DNA (new) : Zycov-d
- Nasal (new)


1
Spike protein
Spike protein genetic material extracted
COVID-19 virus
Genetic material inserted into inactive (harmless) virus
Unrelated, harmless (vector) virus
2
Viral vector vaccine
3
Harmless virus enters cell
Human cell
4
COVID-19 virus spike protein created
COVID-19 virus spike protein
5
Antibodies
Spike proteins are recognized by the immune system, which produces specific antibodies against the COVID-19 virus
6
Immunity
If you're infected with the COVID-19 virus, antibodies bind to virus & stop it from replicating


1
mRNA with instructions for making the spike protein is developed in a lab
2
3
Spike protein
COVID-19 virus
mRNA vaccine
mRNA enters the cell
Human cell
4
COVID-19 virus spike protein created
COVID-19 virus spike protein
5
Antibodies
Spike proteins are recognized by the immune system, which produces specific antibodies against the COVID-19 virus
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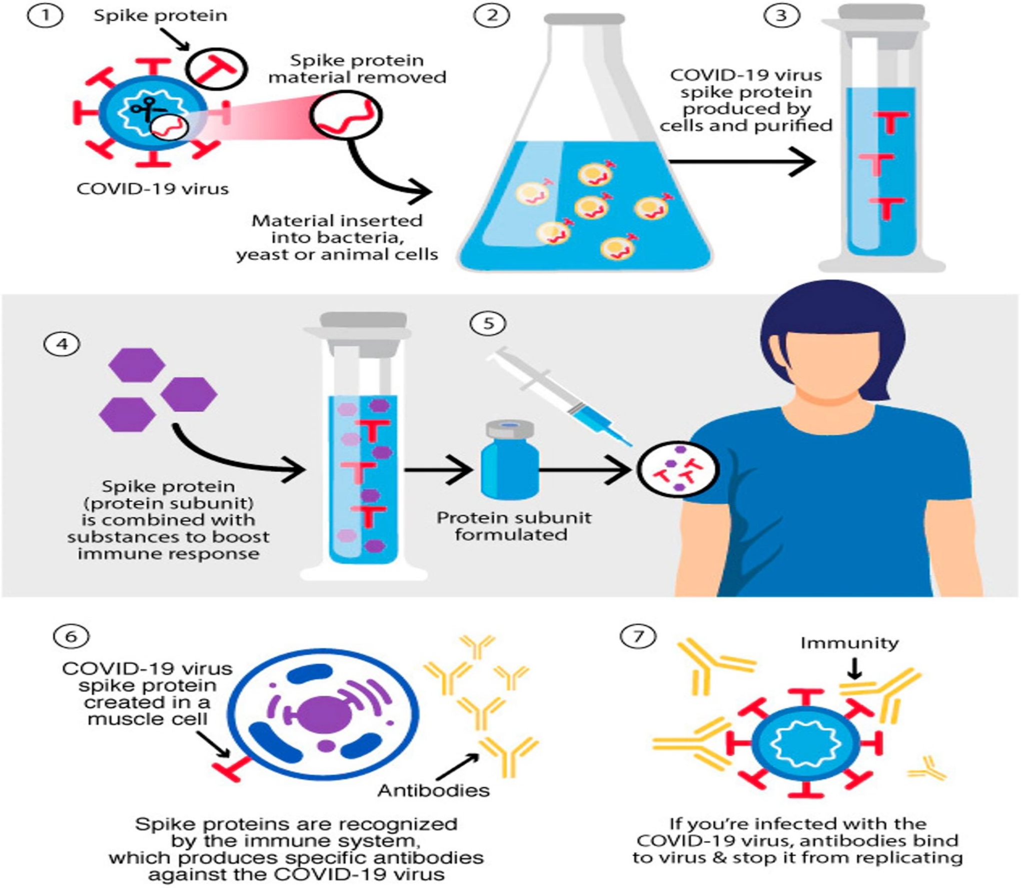

1
Spike protein
Spike protein material removed
COVID-19 virus
Material inserted into bacteria, yeast or animal cells
2
3
COVID-19 virus spike protein produced by cells and purified
4
Spike protein (protein subunit) is combined with substances to boost immune response
5
Protein subunit formulated
6
COVID-19 virus spike protein created in a muscle cell
Antibodies
Spike proteins are recognized by the immune system, which produces specific antibodies against the COVID-19 virus
7
Immunity
If you're infected with the COVID-19 virus, antibodies bind to virus & stop it from replicating

Covid 19 Vaccines in India

- Covishield : Oxford and Astra Zeneca : viral vector vaccine
- In this type of vaccine, genetic material from the COVID-19 virus is placed in a modified version of a different virus (viral vector). When the viral vector gets into your cells, it delivers genetic material from the COVID-19 virus that gives your cells instructions to make copies of the S protein. Once your cells display the Spike proteins on their surfaces, your immune system responds by creating antibodies and defensive white blood cells. If you later become infected with the COVID-19 virus, the antibodies will fight the virus.

Covid 19 Vaccines India

- Covaxin : bharat biotech
- Inactivated virus technology
- Sputnik V
- Similar to covishield
- Zycov – d
- Inocovacc

Zycov -D

- Developed by Zydus Cadilla, got emergency use authorization
- World's first DNA vaccine against covid 19, it has been granted emergency use authorization.
- Developed in partnership with DBT under Mission Covid Suraksha
- Can be administered without a needle : special intra dermal system called tropis that uses precise stream of jet to penetrate the skin.
- For adults and those above 12 years old
- 3 doses needed.

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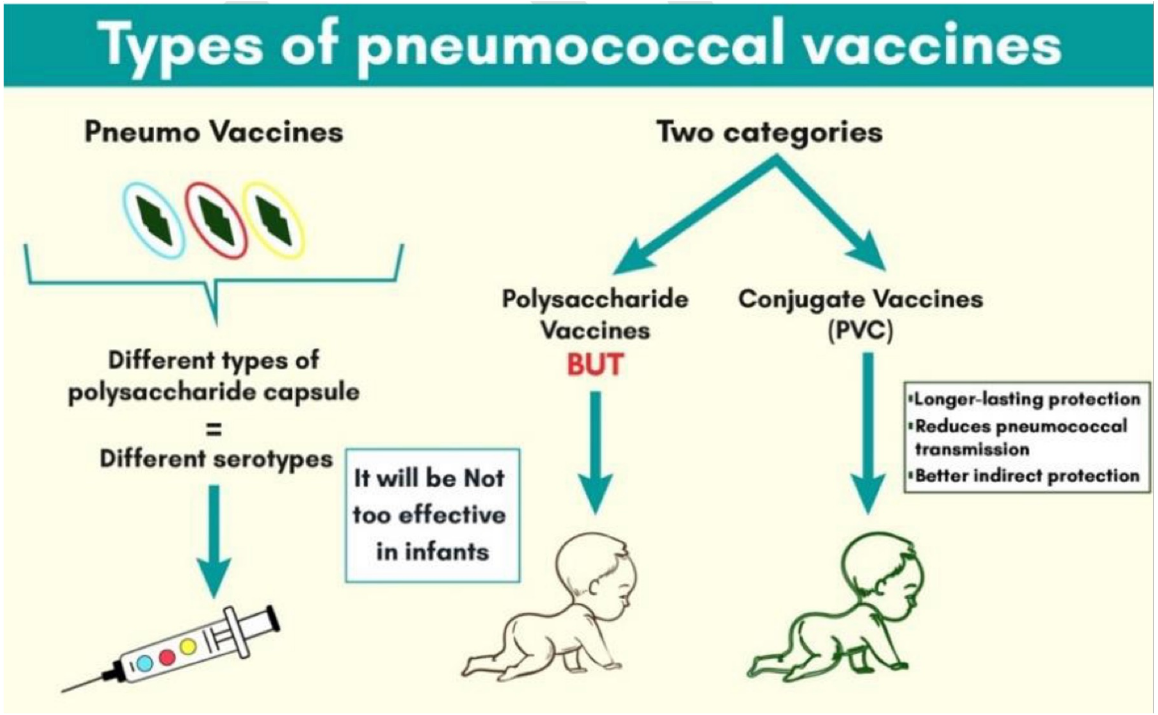
iNCOVACC

- Bharat biotech has developed it as the world’s first intranasal vaccine for Covid-19.
- Received emergency use authorization.
- Adenovirus vector based vaccine.

Pneumococcal Vaccine

- Pneumococcal Conjugate vaccine has been included under Universal Immunisation Programme.
- UIP : 12 Vaccine preventable diseases : Diphtheria, Pertussis, Tetanus, Polio, Measles, Rubella, severe form of Childhood Tuberculosis, Rotavirus diarrhoea, Hepatitis B and Meningitis & Pneumonia caused by Haemophilus Influenzae type B (10 national) and Pneumococcal Pneumonia (Bihar, himachal) and Japanese Encephalitis (endemic districts)
- PCV : provides protection against Streptococcus Pneumonia, which causes pneumonia, meningitis, sinusitis, bronchitis and middle ear infection.


Types of pneumococcal vaccines
Pneumo Vaccines
Different types of polysaccharide capsule
=
Different serotypes
It will be Not too effective in infants
Two categories
Polysaccharide Vaccines
BUT
Conjugate Vaccines (PVC)
Longer-lasting protection
Reduces pneumococcal transmission
Better indirect protection

PCV

- If only polysaccharide is present (bacterial capsule) : the immunogenicity is less.
- Carrier protein is attached to make the immunogenicity higher : stronger immune response.

Polysaccharide (weak antigen) + Protein (Strong Antigen)

↓

Conjugate.

BCG vaccines (Bacillus Calmette Guerin)

- It has been 100 years since the BCG vaccine was first used in humans in 1921 against tuberculosis (TB). Only licensed vaccine available.
- Helps prevent severe childhood tuberculosis.
- Mycobacterium Tuberculosis is the causative organism, transmission is through respiratory pathway.
- Efficacy is higher farther from the equator because tropical regions have many environmental strains of Mycobacterium that lead to interference with vaccine action.
- Also used for Other mycobacterial diseases like leprosy, Buruli ulcer and immunotherapy against urinary bladder cancer and malignant melanoma.
- TB in India : India has 27% of global disease burden.
- ICMR : trials of recombinant BCG (vpm1002) and Mycobacterium Indicus Pranii

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Toxoid Vaccine

- A toxoid is an inactivated toxin (usually an exotoxin) whose toxicity has been suppressed either by chemical (formalin) or heat treatment, while other properties, typically immunogenicity, are maintained. Toxins are secreted by bacteria, whereas toxoids are altered form of toxins; toxoids are not secreted by bacteria. Thus, when used during vaccination, an immune response is mounted and immunological memory is formed against the molecular markers of the toxoid without resulting in toxin-induced illness. Such a preparation is also known as an anatoxin.
- Example : Diphtheria Toxoid Vaccine

Tetanus Vaccine

- Active Tetanus Vaccination : Tetanus Toxoid : Given during childhood
- Also given to those who have an injury with uncertain immunization history
- Passive Vaccination : high risk : given to those who don’t have immunity and need immediate help

Test of 4 Questions!

This is not a learning exercise. You must get all of these correct.

Cut off : 4/4

Question 1 (Previous Year Question 2020)

Q. What is the importance of using Pneumococcal Conjugate Vaccines in India?

1. These vaccines are effective against pneumonia as well as meningitis and sepsis.
2. Dependence on antibiotics that are not effective against drug-resistant bacteria can be reduced.
3. These vaccines have no side effects and cause no allergic reactions.

Which of the statements given above is/are correct?

A. 1 only
B. 1 and 2 only
C. 3 only
D. All of the above

Question 2 (Previous Year Question 2021)

Q. With reference to recent developments regarding ‘Recombinant Vector Vaccines’, consider the following statements:

1. Genetic engineering is applied in the development of these vaccines.
2. Bacteria and viruses are used as vectors.

Which of the statements given above is/are correct?

A. 1 only
B. 2 only
C. Both
D. none

Question 3 (Previous Year Question 2022)

In the context of vaccines manufactured to prevent COVID-19 pandemic, consider the following statements :

1. The Serum Institute of India produced COVID-19 vaccine named Covishield using mRNA platform.
2. Sputnik V vaccine is manufactured using vector based platform.
3. COVAXIN is an inactivated pathogen based vaccine.

Which of the statements given above are correct?

A. 1 and 2 only
B. 1 and 3 only
C. 2 and 3 only
D. All 1, 2 and 3

Question 4 (Previous Year Question 2022)

Q.10 – Which one of the following statements best describes the role of B cells and T cells in the human body ?

(a) They protect the body from environmental allergens.
(b) They alleviate the body's pain and inflammation.
(c) They act as immunosuppressants in the body.
(d) They protect the body from the diseases caused by pathogens.

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Congratulations on giving the test!

Let’s Discuss The Answers.

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Your Own Mentor

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Answer: B

- Pneumococcal Conjugate Vaccines (PCVs) prevent pneumococcal diseases. Pneumococcal disease refers to any illness caused by pneumococcal bacteria.Streptococcus pneumoniae (pneumococcus) is a leading cause of bacterial pneumonia, meningitis and sepsis in children.
- PCVs could potentially prevent a substantial proportion of episodes of bacteremic disease, pneumonia, meningitis, sepsis and otitis media, especially in young children. Hence, statement 1 is correct.
- The growing resistance of pneumococcal bacteria to commonly used antibiotics underlines the urgent need for vaccines to be used to control pneumococcal disease. PCV prevents antibiotic resistant pneumococcal infections. Hence, statement 2 is correct.
- Redness, swelling, pain, or tenderness where the shot is given, and fever, loss of appetite, fussiness (irritability), feeling tired, headache and chills can happen. Hence, statement 3 is not correct.
- Therefore, option (b) is the correct answer.

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Which of the statements given above is/are correct?

A. 1 only
B. 2 only
C. Both
D. none

Answer: C

● Recombinant vector vaccines are made through genetic engineering. The gene that creates the protein for a bacteria or virus is isolated and placed inside another cell's genes. Hence, statement 1 is correct.

● Numerous viral vectors are available for vaccine development, such as vaccinia, modified vaccinia virus Ankara, adenovirus, adeno-associated virus, retrovirus/lentivirus, alphavirus, herpes virus, etc. Hence, statement 2 is correct.

● Therefore, option (c) is the correct answer.

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B. 1 and 3 only
C. 2 and 3 only
D. All 1, 2 and 3

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Answer: C

- Covisheild is a recombinant, replication-deficient chimpanzee adenovirus vector encoding the SARS-CoV-2 Spike (S) glycoprotein. Following administration, the genetic material of part of corona virus is expressed which stimulates an immune response.
- Sputnik V vaccine is based on a well-studied human adenovirus vector platform.
- COVAXIN is developed using Whole-Virion Inactivated Vero Cell derived platform technology. Inactivated vaccines contain dead virus, incapable of infecting people but still able to instruct the immune system to mount a defensive reaction against an infection.

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Question 4 (Previous Year Question 2022)

Q.10 – Which one of the following statements best describes the role of B cells and T cells in the human body ?

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(b) They alleviate the body’s pain and inflammation.
(c) They act as immunosuppressants in the body.
(d) They protect the body from the diseases caused by pathogens.

Answer: D

- T cells protect us from infection. In our daily lives, we’re constantly exposed to pathogens, such as bacteria, viruses and fungi. Without T lymphocytes, also called T cells, every exposure could be life-threatening.
- B cells create antibodies. B lymphocytes, also called B cells, create a type of protein called an antibody. These antibodies bind to pathogens or to foreign substances, such as toxins, to neutralize them. For example, an antibody can bind to a virus, which prevents it from entering a normal cell and causing infection.

Concept of Mutations

- A mutation is a change in the DNA sequence of an organism. Mutations can result from errors in DNA replication during cell division, exposure to mutagens or a viral infection
- Change in DNA = New genes = Can be beneficial or harmful
- Happens in normal process of cellular repair
- Hiroshima / Nagasaki : Mutagens
- All living cells can mutate

Gene Therapy

- A child is born with genetic defects. One important gene that he has is non functional. What do we do?
- Delivery of normal genes through baby with gene therapy.
- Gene therapy is a collection of methods that allows correction of a gene defect that has been diagnosed in a child/embryo.
- Here genes are inserted into a person's cells and tissues to treat a disease.
- Correction of a genetic defect involves delivery of a normal gene into the individual or embryo to take over the function of and compensate for the non-functional gene.

Gene Therapy

- Biotechnological method to replace defective gene with correct functioning gene.
- 2 types : Somatic cell gene therapy and Germline Gene therapy
- Somatic Gene therapy : Genetic Disorders in individuals can be cured.
- Ex : Haemophilia, Sickle Cell Anaemia (important this year : WHY?)
- Germ Cell Gene therapy : Offspring (child) can be prevented from getting disease.
- Can be used to fight cancer as well : CAR-T cell therapy.

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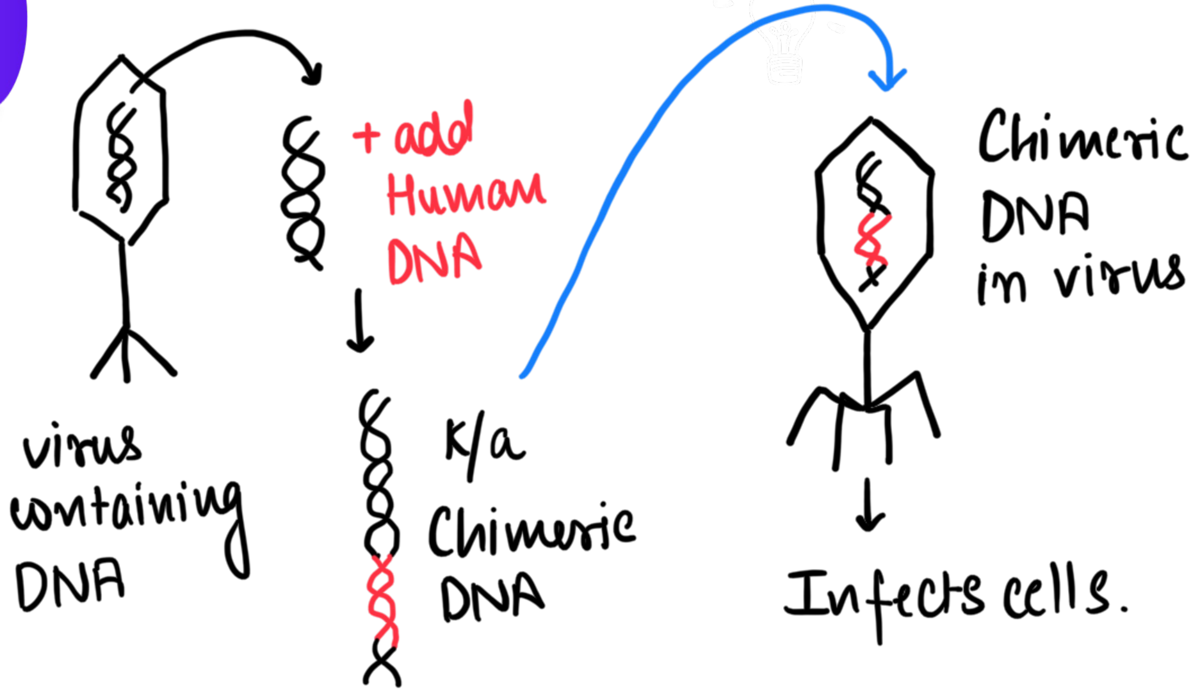
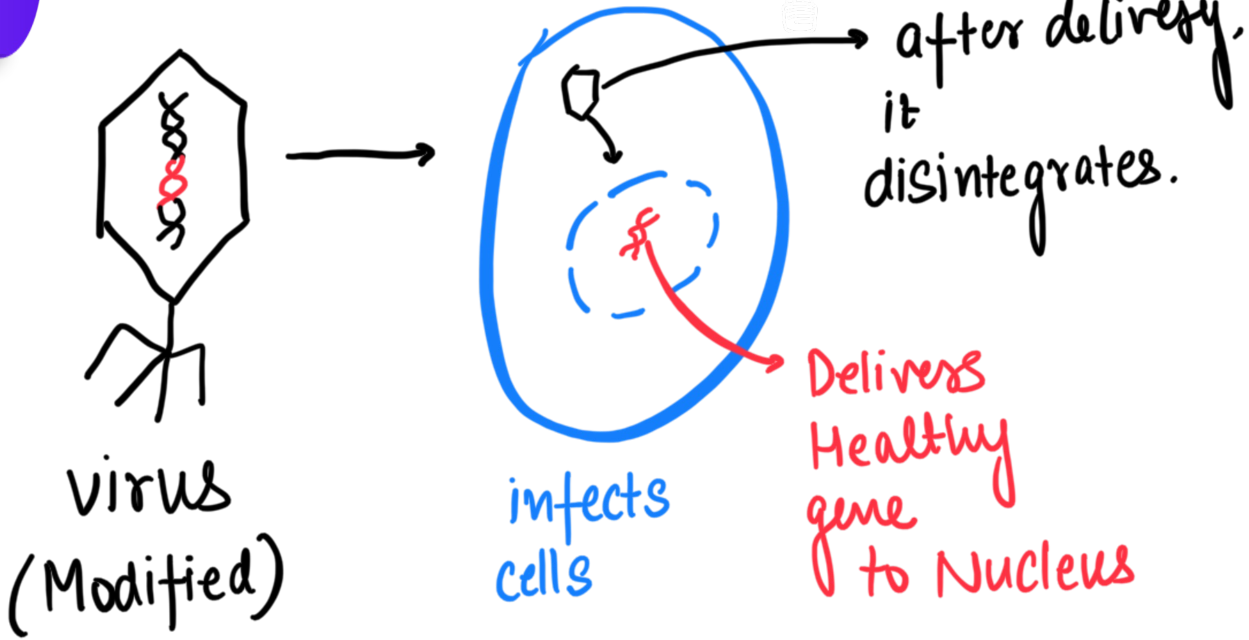
How is it done ?



How is it done (2)



LET'S SEE A VIDEO ON GENE THERAPY

- click here (nationwide children SMA type 1)
- click here for video 2

Gene Therapy

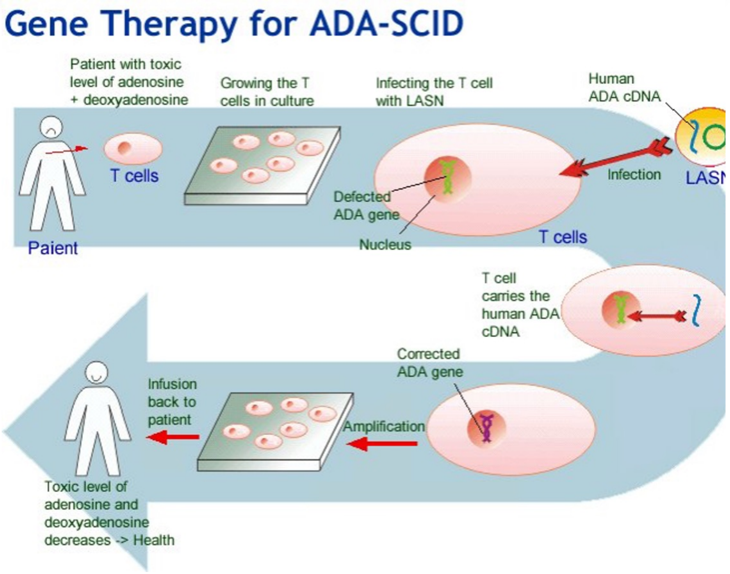
Can be used to :

- 1. Replace bad gene with healthy gene
- 2. Add a new gene to perform function of missing / non functional gene
- 3. Turn off a problem gene.
- Can be done inside the body (using virus) called in vivo
- We are using modified human virus, for example adenovirus.
- Can be done outside the body : in vitro

In vitro gene therapy : Example : ADA - SCID

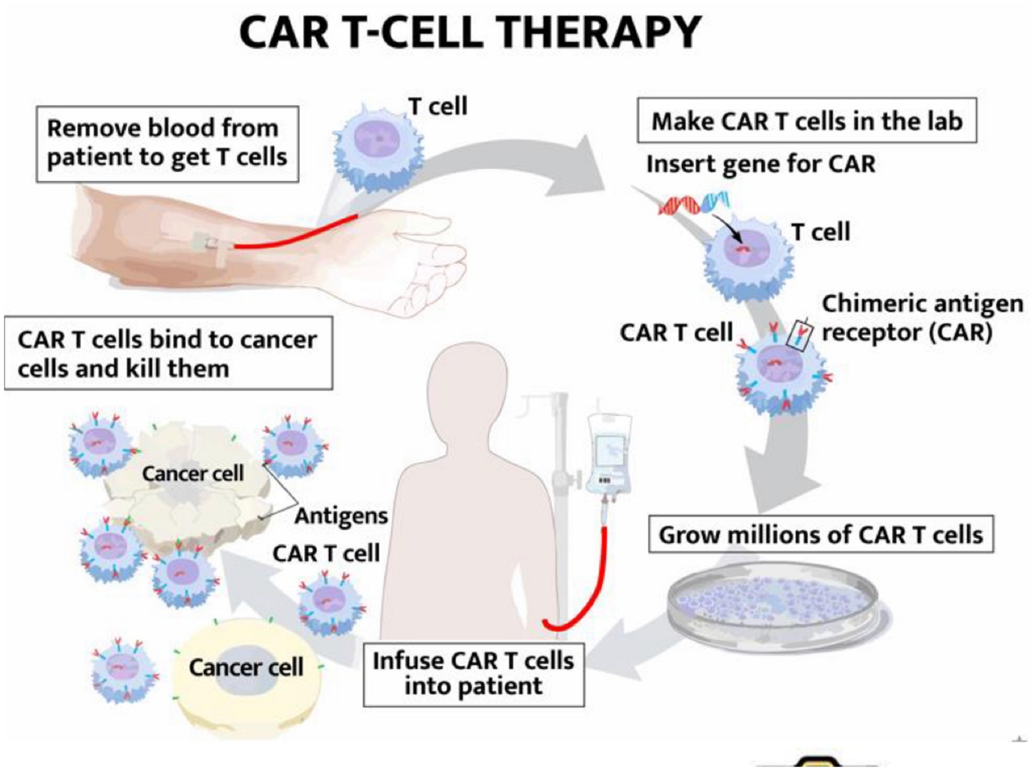
- Adenosine Deaminase Deficiency leading to severe combined immunodeficiency
- T cells of the body need ADA.
- When ADA is absent, patient develops SCID.
- We need to deliver ADA to patient's T cells.
- We use functional ADA gene inserted with viral vector. But virus has to be genetically edited to be made safe.
- T cells are first removed from the patient, edited with viral vector containing new gene, then sent back into the patient.

Gene therapy for ADA- SCID



CAR-T Cell Therapy

- Immune T cells to fight cancer by editing them in the Lab.
- T cells are taken from the patient's blood and are edited by adding a gene for man made receptor : CAR (chimeric antigen receptor)
- CAR is a receptor that binds to certain proteins on cancer cells.
- CAR added T cells attack cancer cells.


CAR T-CELL THERAPY
Remove blood from patient to get T cells
T cell
Make CAR T cells in the lab
Insert gene for CAR
T cell
Chimeric antigen receptor (CAR)
CAR T cell
Grow millions of CAR T cells
Infuse CAR T cells into patient
CAR T cells bind to cancer cells and kill them
Cancer cell
Antigens
CAR T cell
Cancer cell

CAR-T Cell Therapy : Benefits

- Can cure certain types of cancers.
- Unlike chemotherapy (current method of treating cancer) CAR-T cell therapy is used only once.
- Shorter Treatment
- Faster Recovery

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Issues of Gene Therapy

- Short Lived Nature : multiple rounds needed : example : ADA SCID
- Possible solution : Do at early stage.
- Immune response possible, other side effects possible.
- Cannot be used to correct chromosomal disorders, only disorders with single bad gene. (explained later)
- Virus alteration and manipulation : Containment

Issues of Gene Therapy

- Short Lived Nature : multiple rounds needed : example : ADA SCID
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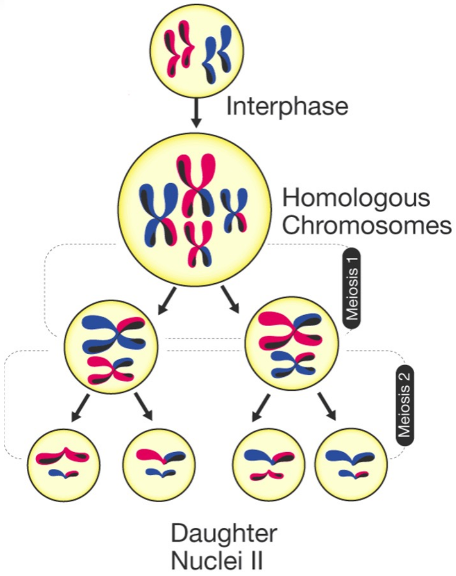
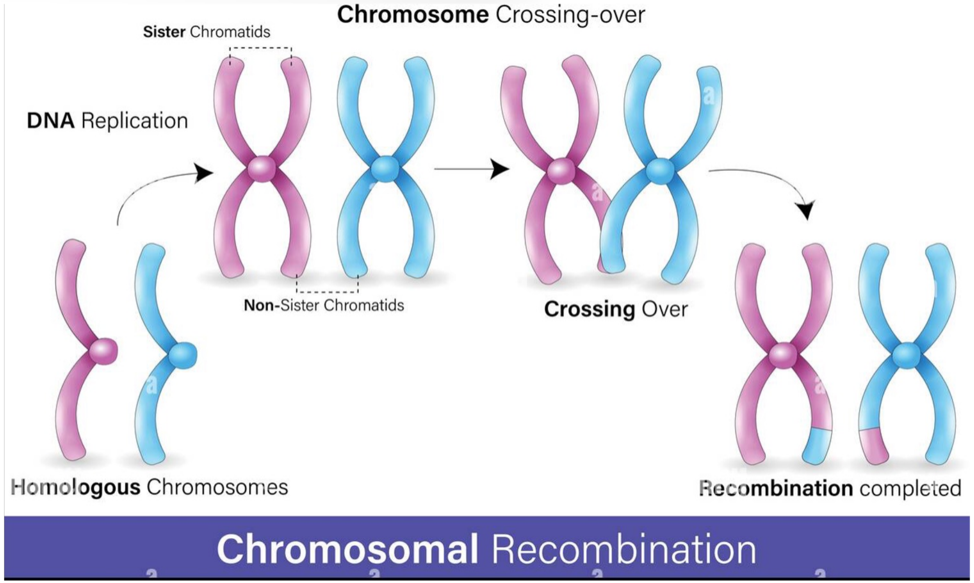
Chromosomal Disorders : defects during mieosis





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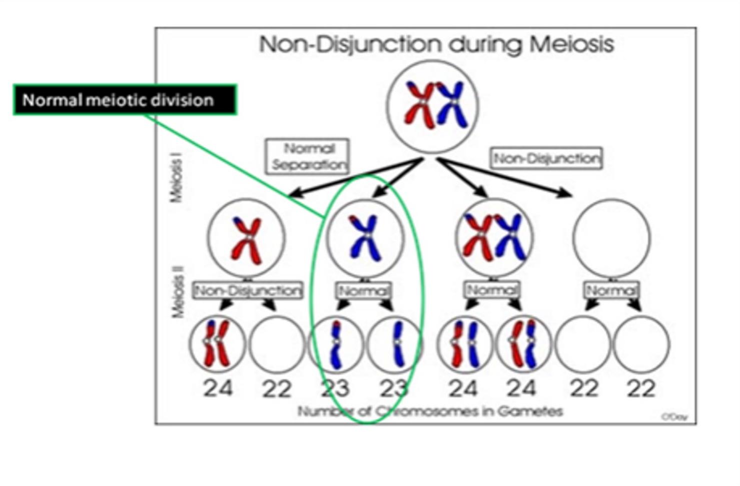
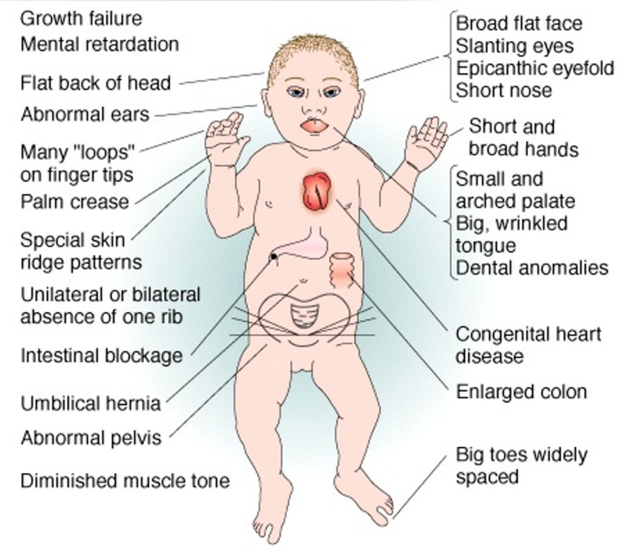
Chromosomal disorders : Down's Syndrome

- Patient has 3 copies of Chromosome number 21
- Leads to severe genetic defects.
- This cannot be cured by gene therapy.



Down’s Syndrome : treat them with sympathy



(a)

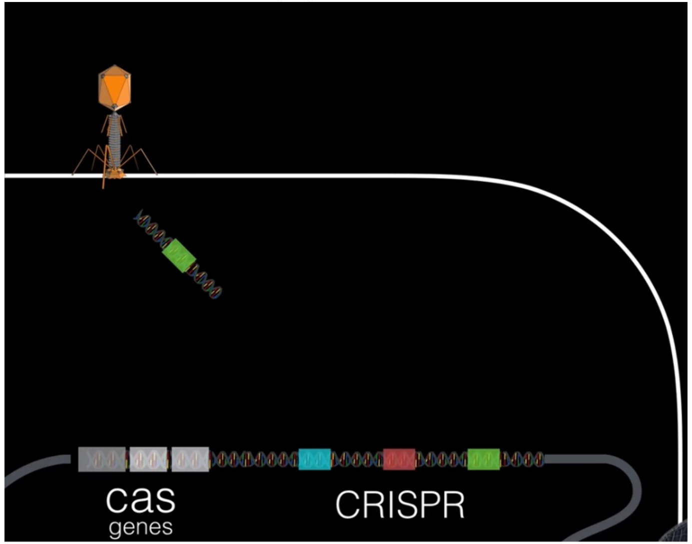
Genome Editing Techniques

- Crispr Cas-9
- Zinc Finger Nuclease
- Talens
- Site Directed Nuclease
- Homing Endonuclease

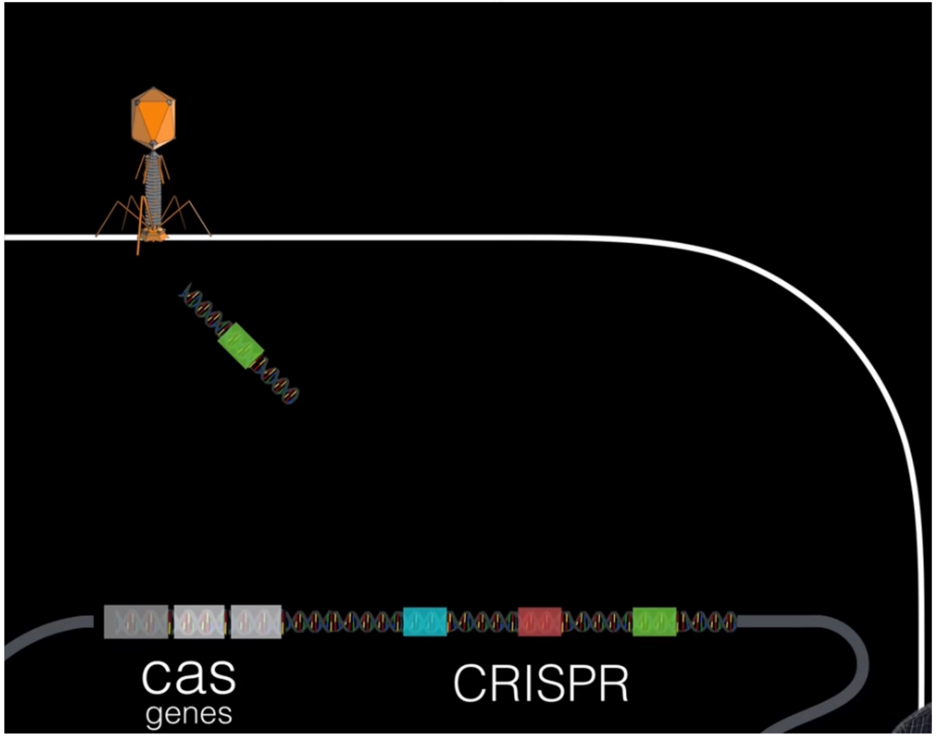
Crispr Cas – 9 and Gene Editing

- What are genes?
- Gene Editing : changing genes in humans
- CRISPR : Clustered Regularly Interspaced short Palindromic Repeats
- CAS : crispr associated protein -9
- Originally found in bacteria
- Molecular level scissors and glue (cut and paste mechanism)

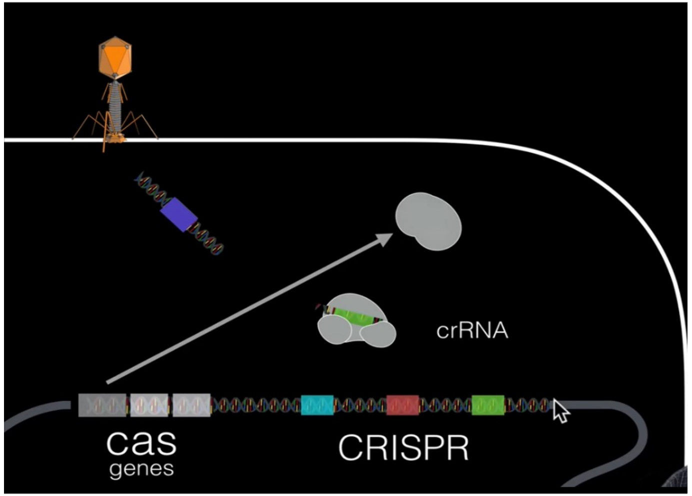
- Palindromes interspaced by spacer DNA
- Cas gene : Cas proteins :
  Helicases (unwind DNA)
  Nucleases (Cut DNA)
- It is a defence system that bacteria uses to defend against viruses that infect bacteria. These viruses are called bacteriophage.

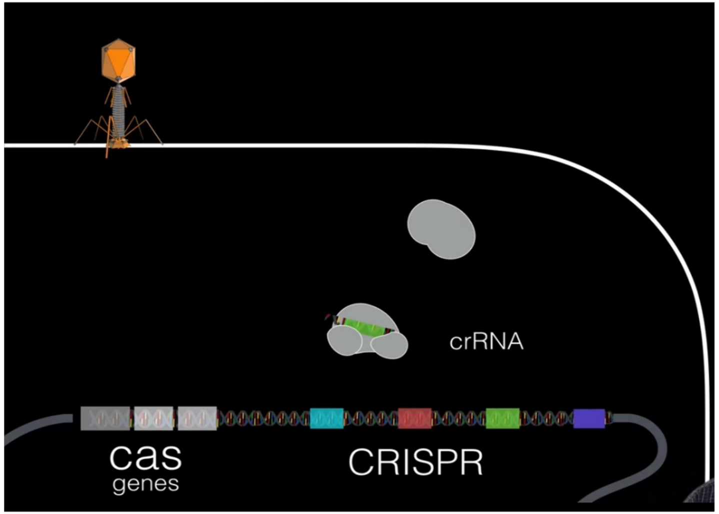


- It is a defence system that bacteria uses to defend against viruses that infect bacteria. These viruses are called bacteriophage.



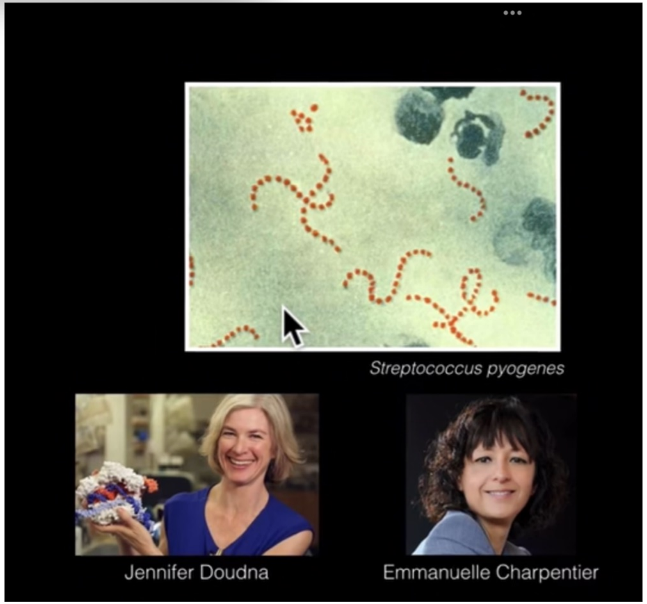
Crispr Cas-9 : Working explained





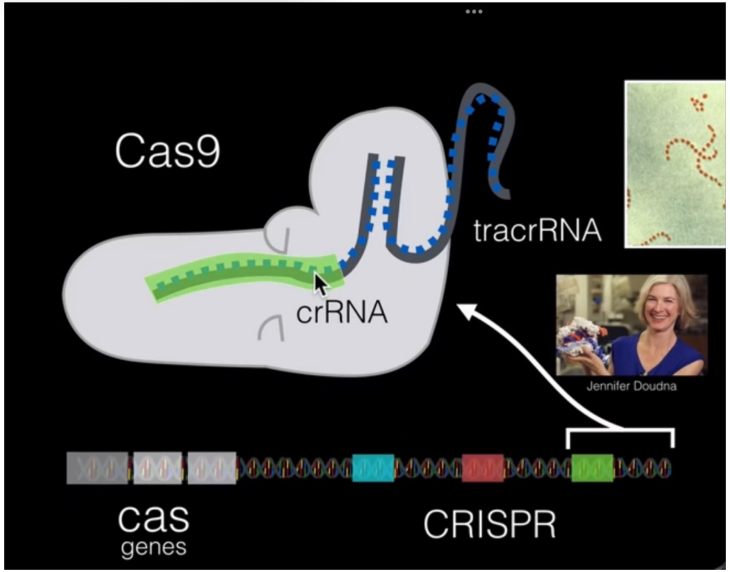
Crispr

- Spacer – repeat – Spacer– Repeat
- Similar to our immune system.
- Scientists thought of a new use for this in genetic editing.

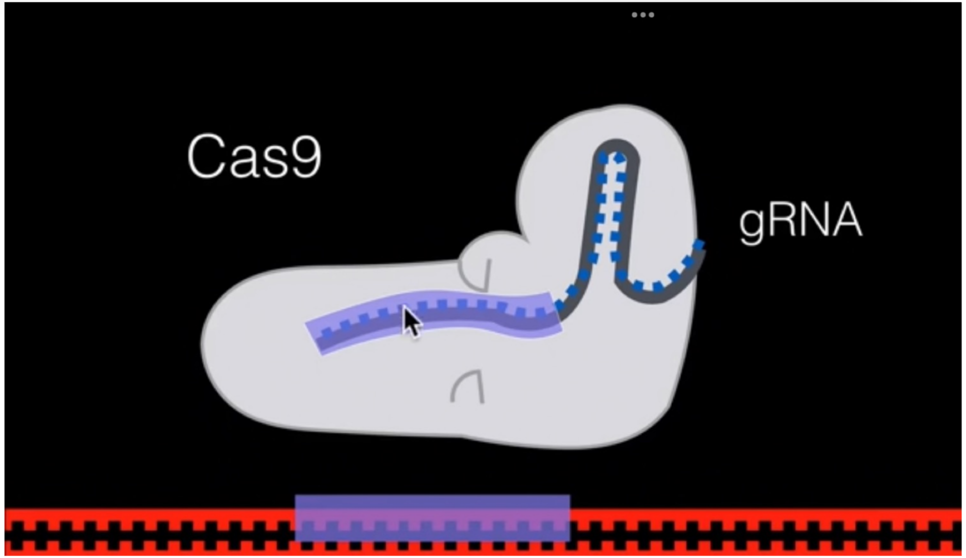


Crispr in genetic editing

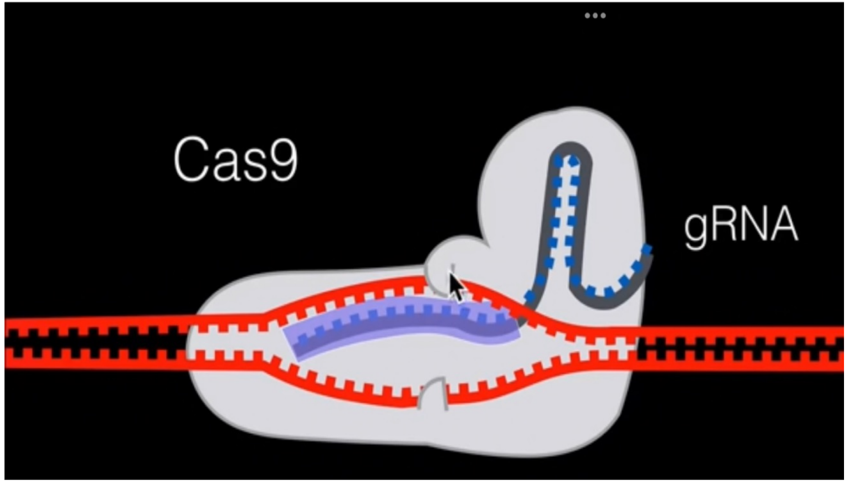
- Nuclease : enzyme responsible for cutting DNA

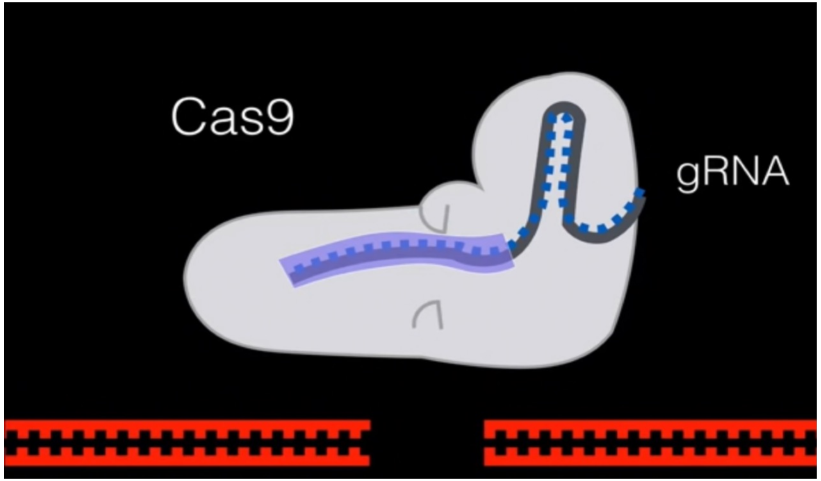


We modified CRISPR to work in our cells.



We modified CRISPR to work in our cells.





CRISPR Cas 9

- Consists of single guide RNA (sgRNA) : complementary to the part of DNA to be cut
- Cas 9 : crispr associated protein 9 : cuts specific part of dna
- It is like using Microsoft word for a document
- Ctrl + F = Find (SgRNA)
- Replace = Delete and change (Cas-9)
- Crispr cas 9 is molecular scissors

Possible Applications

- Gene editing and gene therapy
- Creation of Genetically Modified Organisms
- Designer babies : Chinese Scientist Jailed for this
- Germline editing of reproductive cells

Issues

- Increased risk of mutations and cancer
- Ethical concerns and designer embryos
- Impact on human variation
- Abuse of gene editing
- Safety and off target effects
- Informed Consent
- Justice and equity

Precision Guided Sterile Insects

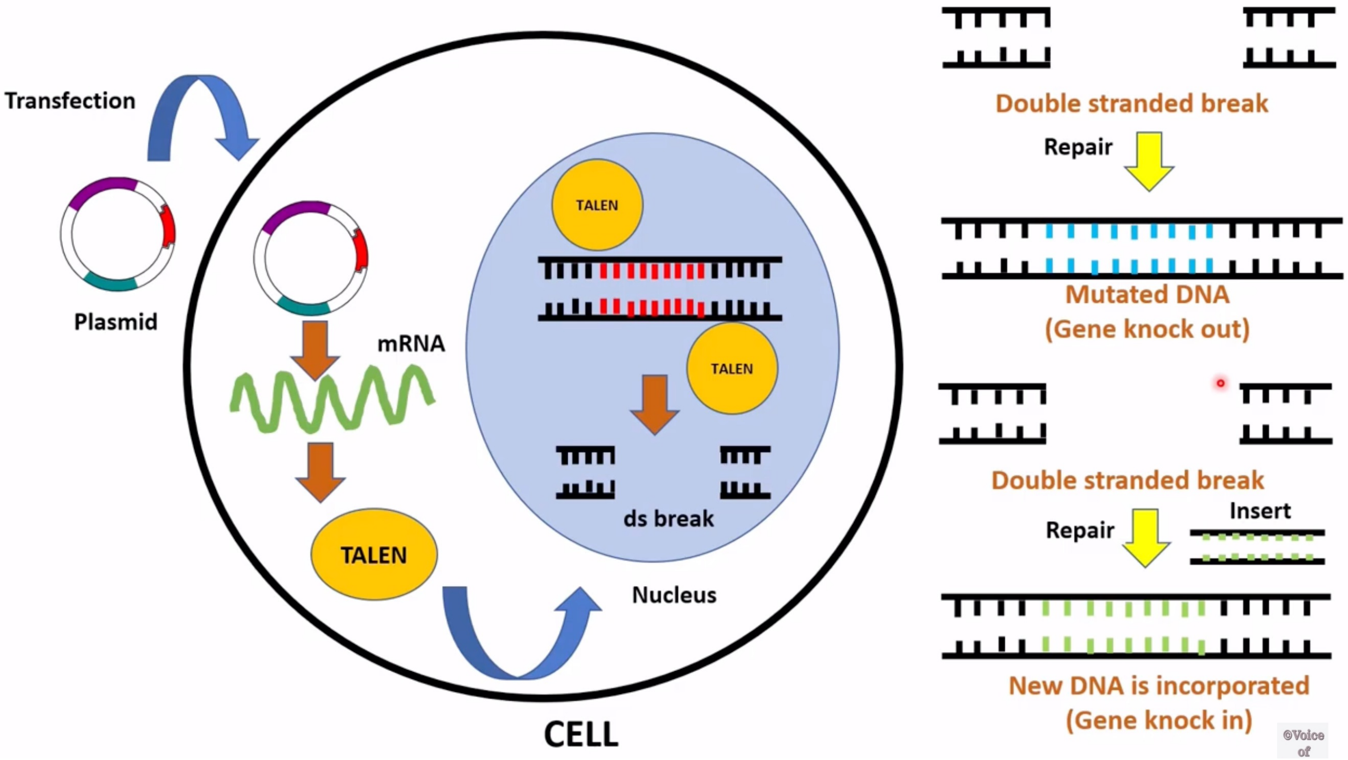
- CRISPR-based system was developed to safely restrain mosquito vectors via sterilization.
- precision-guided sterile insect technique (pgSIT) was used to alter genes in Aedes aegypti, the species responsible for spreading malaria, dengue, chikungunya and Zika.
- pgSIT uses CRISPR to sterilise male mosquitoes and render female mosquitoes (which spread disease) flightless.
- "Gene drive" systems could suppress disease vectors by passing desired genetic alterations indefinitely from one generation to the next.
- Unlike 'gene drive' the pgSITsystem is self-limiting and is not predicted to persist or spread in the environment.

Zinc finger nuclease

- DNA Scissors. This is all you actually need to remember.
- DNA binding part is zinc finger protein. It will recognise portion of DNA to be cut.
- Nuclease part of ZFN (required to cut) is FOK1 Nuclease. (easy to remember? 😂)
- Success rate is low, 1-20%
- Older Technology than Crispr cas – 9.

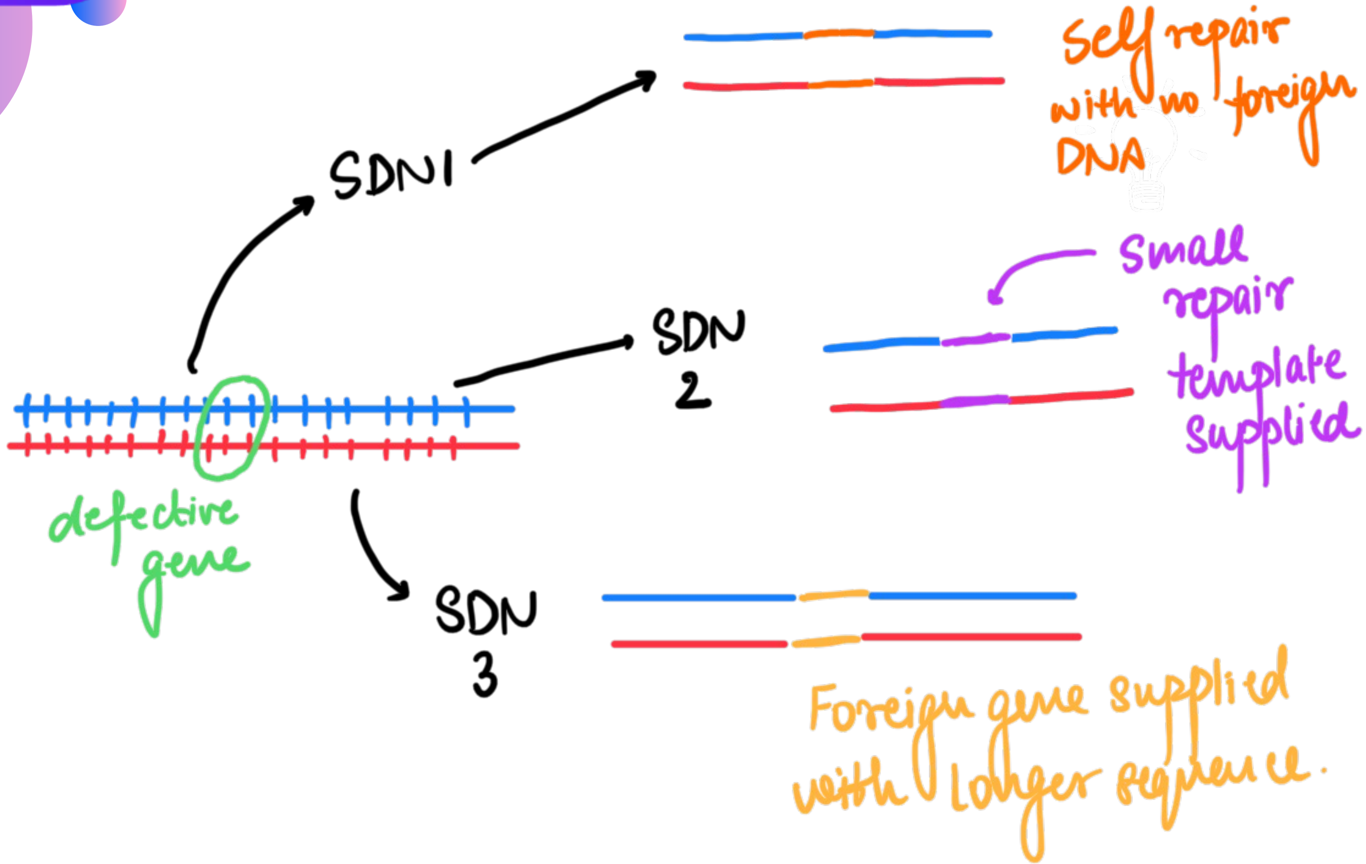
TALENs

- DNA binding domain of TALENs is made of Transcription activator like effector (TALE) domain
- Nuclease Part is same, Fok1 nuclease.
- Also DNA scissors.
- Taken from xanthomonas bacteria.


Transfection
Plasmid
mRNA
TALEN
TALEN
TALEN
ds break
Nucleus
CELL
Double stranded break
Repair
Mutated DNA
(Gene knock out)
Double stranded break
Repair
Insert
New DNA is incorporated
(Gene knock in)

Site Directed Nuclease

- Site Directed Nuclease are genome editing techniques
- All 3 have different functions. (next slide)
- SDN refers to cleaving DNA strands to affect subsequent genome editing, taking advantage of natural repair mechanism to introduce small changes.
- Main SDN technologies : ZFN and TALENs


SDN1
Self repair with no foreign DNA
SDN 2
Small repair template supplied
SDN 3
Foreign gene supplied with longer sequence.
defective gene

Site Directed Nuclease

- SDN 1 and SDN 2 : No alien genetic material is being introduced.
- Result is indistinguishable from conventionally bred crop varieties.
- SDN 3 : Genes of Foreign origin are introduced.
- SDN 1 and SDN 2 : have been exempted from 1989 rules on GM crops.
- SDN 3 : Will come under the rules if Foreign DNA **exceeding 20 base pairs** in inserted.

Homing Endonuclease / Meganuclease

- Another type of molecular scissors
- Cuts the DNA and then you can choose to add or not add any gene.

Gene Doping

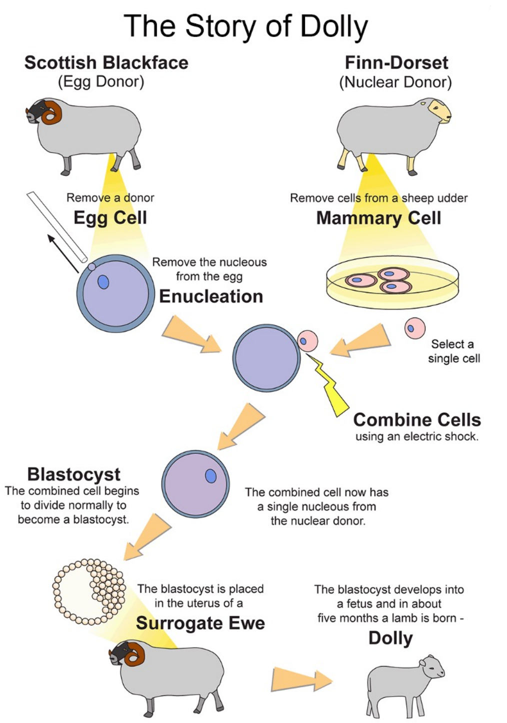
- Changing DNA to enhance performance by athletes
- Ex vivo (in vitro)
- In vivo
- WADA issues

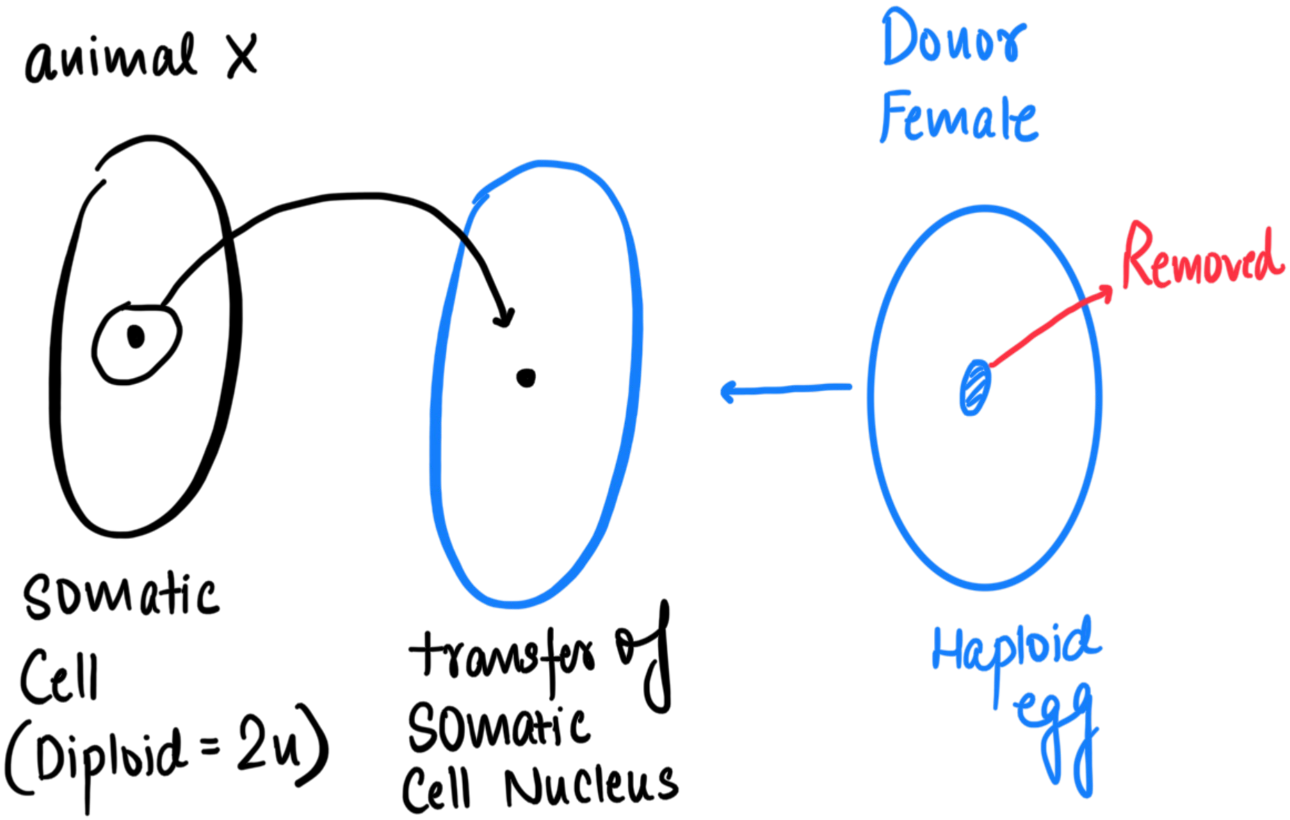
Animal Cloning

- Identical twins : from 2 parents
- Cloning is the development of cloned animal from a single parent. Also known as Reproductive Cloning.
- Method : Somatic Cell Nuclear Transfer Technique

Animal Cloning and Somatic Cell Transfer

- Cloning : exact same genes of progeny
- Dolly the sheep : Ian Wilmut
- Reproductive cloning
- We will study **<u>Therapeutic Cloning after stem cells</u>**




animal X
Somatic
Cell
(Diploid = 2n)
transfer of
Somatic
Cell Nucleus
Donor
Female
Removed
Haploid
egg

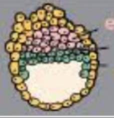
Cloning

- Recently China has cloned a wild arctic wolf first time in the world
- Clone is NOT completely identical : does not always look exactly the same as environment also plays a role in determining physical features of an individual.
- 3 types of artificial cloning :
- Gene/DNA cloning : DNA fragment from one organism to a self replicating genetic element, such as bacterial plasmid.
- Reproductive Cloning : Transferring nuclear material isolated from somatic cell into enucleated egg.
- Therapeutic cloning.

Stem Cells

- Embryo
  - Totipotent (zygote)
  - Pluripotent (early cell division)
- Adult
  - ↓ Multipotent
  - Specific to organs.
  - eg: Bone Marrow
- induced pluripotent
  - Normal Somatic cell induced to form Stem Cells.

Stem Cells

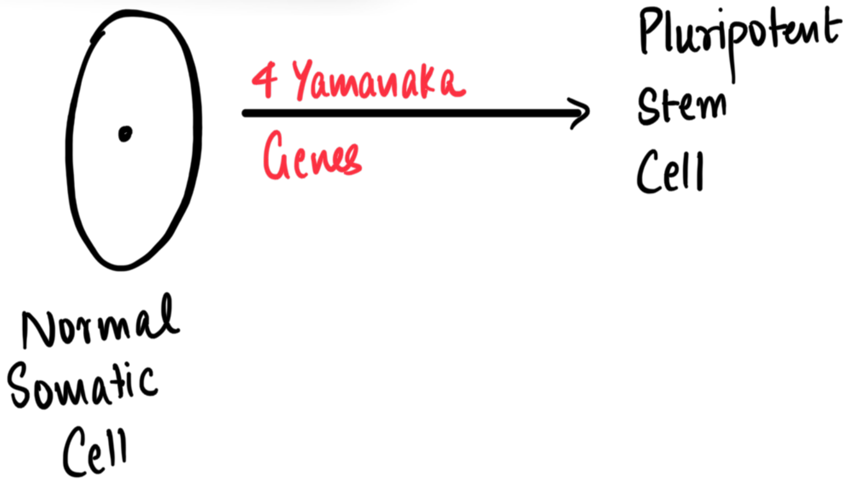
STAGE OF EMBRYOGENESIS*	EVENTS	IMAGES (NOT TO SCALE)	TOTIPOTENT OR PLURIPOTENT?
Fertilization Carnegie Stage 1	Ovum and sperm fuse to form a diploid zygote.		Totipotent
Cleavage Carnegie Stage 2	The zygote undergoes cleavage via mitosis to increase the number of cells within the zona pellucida.	Cells have a distinctly round shape. 	Totipotent
Compaction** Carnegie Stage 2	Blastomeres form junctions with one another, so the cells lose their round shape. The resulting formation is a mulberry-shaped mass of cells within the zona pellucida called a morula, which appears when there are 16 cells.	 Cell boundaries are less clear.	Totipotent
Differentiation Carnegie Stage 2	Cells are divided into distinct lineages: the inner cell mass (epiblast and hypoblast) and the trophoblast.		**Epiblast = Pluripotent** Hypoblast and trophoblast are neither totipotent nor pluripotent

Clarity for UPSC by Dr. Shivin

Ed Sarrthi Your Own Mentor

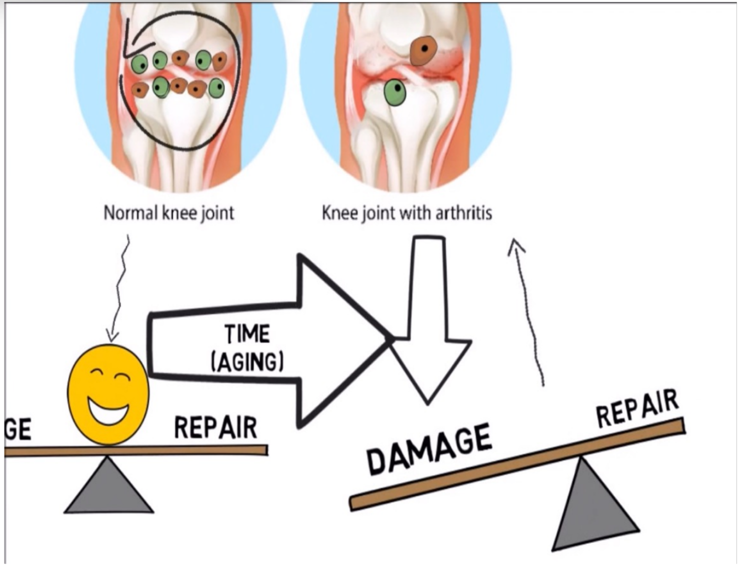
Induced Pluripotent Stem Cells

- Conversion of Somatic Cells to form Pluripotent Stem Cells.
- 4 specific genes (Yamanaka factors) when introduced into normal somatic cell, turn it into stem cell.
- Shinya Yamanaka : Nobel prize in 2012



Stem Cell Therapy

- Osteoarthritis of Knee : degenerative disease
- Where damage overtakes normal repair.
- Here, we can use patient’s own stem cells to provide
- Regenerative support and relieve the injuries an The pain.
- Therefore, Stem Cells are a form of regenerative medicine



Stem Cell Therapy

- Can be used to replace dead / damaged tissues by normal functioning cells produced by stem cells.
- Can be used for diseases like :
  - Thalassemia
  - Leukemia
  - Osteoarthritis and Osteoporosis
  - Blindness
- Advantages : Minimal Immune rejection because cells are derived from the body itself.

Future prospects

- Body organs can be grown in labs if we combine Stem Cells and 3D printers.
- ICMR : guidelines for stem cells in India.
- Stem Cell Banking : storage of umbilical cord stem cells at -170°C with liquid nitrogen to be used for future therapy.

Challenges

- Synthetic biology : seen as intereference with nature
- Biosafety concerns
- Abuse of Technology
- Genetic Divide
- Ethical Concerns

Therapeutic Cloning

- Producing a healthy replica of a human organ
- Better for transplanting organs : no immunosuppressant
- It is like reproductive cloning till production of embryo. Produced embryo is allowed to grow in lab.
- Embryo is allowed to grow and then **stem cells are extracted**
- Treat diseases by replacing damaged cell
- Applications : Ageing, cancer, implant surgeries.
- Ethical issues : Divine command theory, life begins at conception (Christian ethics); Violative of Natural laws

Applications of RDT / Biotechnology [India]

Green Biotech (agriculture)

(i) Yield
(ii) Biopesticides : BT Brinjal
(iii) HYV → GM Mustard
(iv) Biofortification → golden Rice (vit.A)
(v) Shelf life → Flavr Savr tomato (export)
(vi) disease (Sambha Masuri Rice)
(vii) RNAi → plant vaccination
(viii) cold tolerant Strawberry
(ix) Better Breeds → cows. etc.

Blue Biotechnology
(Marine)

- → aquaculture (sea-food cultivation)
- → Transgenic fish → freeze resistant fish
- → Conservation of species
- → seaweed cultivation → ex: agar-agar.

Red Biotechnology (Health)

- prevention
  - vaccines
  - transgenic plants (Nutrition)
  - transgenic animals → (Study)
- diagnosis
  - MAb
  - DNA probes
  - PCR
  - ELISA
- treatment
  - Human insulin
  - gene therapy
  - Stem cell therapy
  - Recombinant protein (γ-IFN)

Environmental

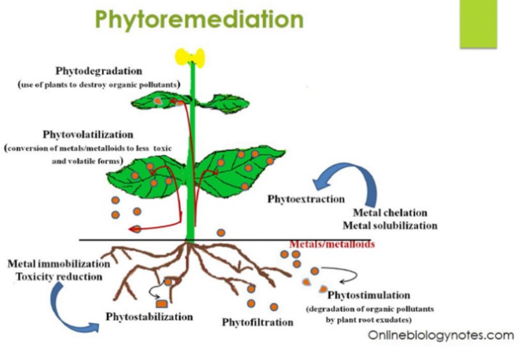
oil
zapper
Bacteria
(Bio Remediation)
oliovorus-S (TERI)

phytoremediation

use of plants and associated soil microbes to reduce toxic elements.

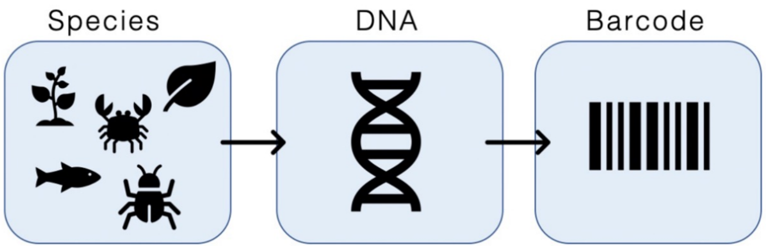
eucalyptus

DNA Barcoding.



DNA Barcoding

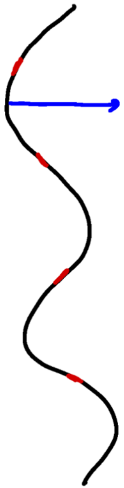
DNA barcoding is a method of species identification using a short section of DNA from a specific gene or genes. The premise of DNA barcoding is that by comparison with a reference library of such DNA sections (also called "sequences"), an individual sequence can be used to uniquely identify an organism to species, just as a supermarket scanner scans barcodes.



Forensic Biotechnology

- DNA fingerprinting : identification of criminal by use of biotechnology.
- Here we use something called variable number of tandem repeats (VNTR)
- Use to develop DNA bands in Gel electrophoresis and identify a person.
- Can be used in :
- Criminal Investigation
- Paternity testing
- Immigration Cases
- Child swapping cases
- Disaster victim identification

DNA fingerprinting



Non coding Region → Short Tandem Repeats → 0.1% different part.

ex : AGAC AGAC AGAC → Number of Repeats are different.

DNA
- → Bulk DNA (99.9%)
- → Satellite DNA (0.1%)
  - → Micro Satellite
  - → Mini Satellite
    - → VNTR (variable No. of Tandem Repeats)

Isolation and extraction

↓

Digestion of DNA by endonuclease

↓

Cut at specific place

↓

Gel electrophoresis → But cant see.

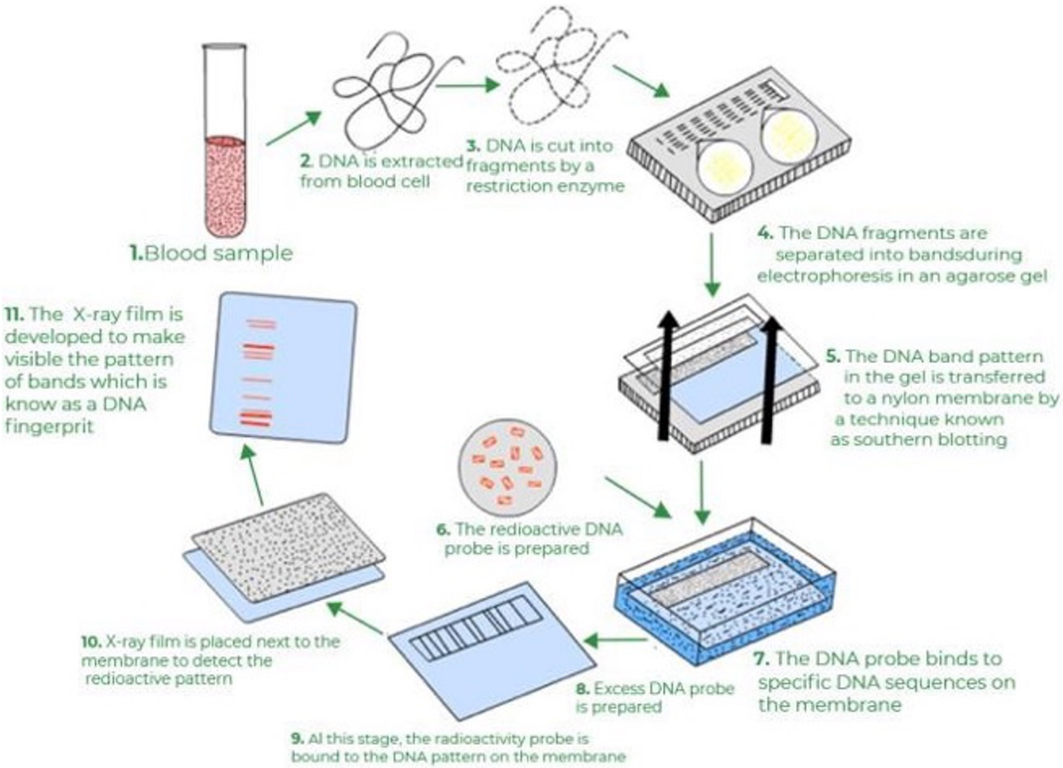
↓

Southern Blotting → Nitrocellulose

Hybridisation with Radioactive probe

↑

autoradiography


1.Blood sample
2. DNA is extracted from blood cell
3. DNA is cut into fragments by a restriction enzyme
4. The DNA fragments are separated into bandsduring electrophoresis in an agarose gel
5. The DNA band pattern in the gel is transferred to a nylon membrane by a technique known as southern blotting
6. The redioactive DNA probe is prepared
7. The DNA probe binds to specific DNA sequences on the membrane
8. Excess DNA probe is prepared
9. Al this stage, the radioactivity probe is bound to the DNA pattern on the membrane
10. X-ray film is placed next to the membrane to detect the redioactive pattern
11. The X-ray film is developed to make visible the pattern of bands which is know as a DNA fingerprit